Improving RAG Answer Quality by Learning to Balance Vector and Graph Retrieval.

by

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Declaration of Authorship

This report, Improving RAG Answer Quality by Learning to Balance Vector and Graph Retrieval., is submitted in partial fulfillment of the requirements of Master of Science in Artificial Intelligence at Munster Technological University Cork. I, William Barrett, declare that this thesis titled, Improving RAG Answer Quality by Learning to Balance Vector and Graph Retrieval.and the work represents substantially the result of my own work except where explicitly indicated in the text. This report may be freely copied and distributed provided the source is explicitly acknowledged. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at Munster Technological University Cork or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this project report is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

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Abstract

Faculty of Engineering and Science
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Master of Science

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Retrieval-Augmented Generation (RAG) dynamically integrates external knowledge into large language model-based (LLM) systems. LLMs perform well for general tasks but hallucinate when met with information they are not trained on. RAG is a proven solution to this but only performs well in specific chunk-based lookups. RAG fails to identify thematic data across an entire corpus. In 2024 Microsoft introduced a graph-based approach named GraphRAG to address this. GraphRAG prebuilds a graph model by identifying entities, communities, and relationships across an entire corpus. Most current research focuses on the integration of vector-based or graph-based RAG in isolation. In research where both are combined, the approach is a fixed equal weighting or selection of which to use. A current gap is that when both are combined, there is little to no research adjusting the split based on the incoming query. This means a question about how two entities relate will get treated the same as a fact lookup. The purpose of this research is to build and evaluate blending strategies to address this gap. These strategies will analyze incoming queries and determine the best means of blending the results of both vector and graph-based RAG. The contribution of this research will include an investigation into multiple classification and blending strategies, combined with an investigation into if adaptive routing outperforms individual retrieval strategies.

Acknowledgements

The acknowledgements and the people to thank go here, don't forget to include your project supervisor...

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Abbreviations

LAH **L**ist **A**bbreviations **H**ere

For/Dedicated to/To my...

Chapter 1

1.1 Research Context

Lewis et al. [1] introduced Retrieval-Augmented Generation (RAG) to enhance large language models (LLMs) with external knowledge through vector based retrieval. Later, Edge et al. [2] highlighted that vector RAG falls short when it comes to global or sense making queries and proposed a graph based approach called GraphRAG. GraphRAG allows the system to answer global queries like 'what are the main themes in this dataset?'. However, GraphRAG has its own limitations. Graph construction introduces an expensive upfront cost, and for simple query-focused summarization, vector RAG is far more efficient. This highlights the need for hybrid approaches to retrieving knowledge outside of parametric memory.

Xu et al. [3] tested this by deploying a hybrid model to production for six months. This hybrid approach was shown to reduce the median per-issue resolution time by 28.6%. However, their system relies on a fallback approach where they try graph search first and on failure it reverts to vector-based retrieval. There is no means to blend both sources and adjust the balance based on the query.

Jeong et al. [4] proposed Adaptive RAG which trains a classifier to classify the query and route between no retrieval, single-step, or multi-step retrieval. This outperformed single step retrieval and used half the compute time when performing multi-step approaches; this proves that classifying queries to pick the right strategy improves retrieval quality. However, this works purely against vector-based RAG and does not consider graph retrieval at all.

Guo et al. [5] used reinforcement learning to fine tune an LLM to route between vector and graph retrieval, but this requires fine tuning the LLM's weights which makes adoptability difficult.

Min et al. [6] combined vector and graph retrieval using RRF [7] and showed a 15% improvement over vector-only baselines. However, their system runs both retrieval strategies for every query with a fixed equal weighting. There is no mechanism to skip graph or vector retrieval when the query doesn't need it.

No current work uses an adaptive balancing approach to blend vector and graph-based retrieval. This research investigates strategies to address this gap and allow better question answering across query types by weighting the retrieval type based on the incoming query.

1.2 Research Aim

The aim of this research is to investigate if adaptive, classification-based blending of vector and graph retrieval can outperform fixed strategies in RAG systems.

1.3 Research Objectives

1. To investigate if rule-based, LLM-based and model-based classifiers can determine retrieval weights between vector and graph to provide a lightweight alternative to fine tuning the LLM directly.
2. To determine if a weighted blending strategy can outperform fixed retrieval strategies that apply the same retrieval approach regardless of query type.
3. To identify which query characteristics should influence the balance between vector and graph retrieval.
4. To determine if a threshold exists to skip a retrieval type entirely, addressing the overhead of running both for every query.

Chapter 2

Literature Review

The key question to answer in this chapter is: "What has been done/is being done".

You must present a summary of background research - what has been done in this area; any related work. Provide reader with information to understand the rest of the report. This chapter comprises around 3000 words and should put your project into context within the field. You must identify the main sources of information you should be aware of within your chosen area and use this activity to strengthen your knowledge in this area. So here you should develop a knowledge of not only your specific topic but also how it is changing.

Include relevant references. For the (rare) projects with no implementation component, e.g., using neural networks to explore or extend the properties of some mathematical model, or to make predictions of some kind, one might break this chapter into two and write a chapter on neural networks and another on the area of application.

To conduct this activity you should aim to read most of a relevant paper every other day for the remainder of the project, make a brief summary of the main findings and how they relate to your work (less than 30 words). You should continue this activity after your project as CPD.

You should begin your journey of discovery in reverse order to the listing above (which is given in order of academic importance/significance). So when you are researching your topic first look up some TedX talks or youtube tutorials, then research what companies are doing in the area, then get a handful of very good texts on the core topics of your area (anything older than 5 years usually is not helpful here) and finally start reading conference or journal papers (again newer is better here). In particular during this section you may need to use tables to list resources. These are also automatically formatted in latex thus allowing you to concentrate on content. for example table 2.1.

Parameter	PET
Youngs Modulus	2800-3100MPa
Tensile Strength	55-75MPa
Glass Temperature	75°C
Density	1400kg/m^3
Thermal Conductivity	0.15-0.24Wm^{-1}K^{-1}
Linear Expansion Coefficient	7×10^-5
Relative Dielectric Constant @ 1MHz	3
Dielectric Breakdown Strength	17kVmm^{-1}

Table 2.1: PET Physical Properties

Make sure you cite[**?**] as many **relevant** papers [**?**], luckily Latex will take care of the ordering of the citations [**?**] for you.

The aim here is that you find the trends in your topic (3), and more in general in the area in which your topic resides (2) your project falls under and from these trends you develop your initial project question further and begin to get insights into how others have solved/approached similar problems. Think of this section as colouring in your initial idea. Before you approach this section you should read at least 4/5 good review papers. Also remember if you find that you are detailing topics that you have not introduced earlier in this section you need to add a reference for background reading.

Chapter 3

Research Objectives

1. Create a hybrid retrieval system that to dymacially weigh and route queryes between vector and graph-based retireval with modifying the underlying Large Language Model.
2. Develop and evaluate rule-based, llm-based, and model-based classifiers to weigh incoming queries between vector and graph retrieval.
3. Evaluate if a weighted blending strategy can outperform fixed retrieval strategies.
4. Analyse which query characteristics should influence the retrieval weighting and if there is a threshold to completely remove a strategy.

Chapter 4

Implementation

This chapter should include the most relevant/important code listing and discuss relevant/interesting aspects; explain features you wish to highlight.

4.1 Important Features

For each aspect you could provide the following info:

1. Description of the difficulty: Brief description of the problem you found.
2. How did it affect the original project design?: Indicate how this difficulty affected:
   (a) the architecture of your solution
   (b) if it represented a risk to your overall project (i.e. was it a FR)
   (c) if it affected your methodology to develop your project (i.e. did developing this aspect change the design)
   (d) if it changed your implementation schedule (i.e. did it delay other items)
   (e) if it changed the evaluation plan (i.e. was more/less testing now required?)
   (f) If there is complex code/configuration details discuss it here also.

From January to September you will be developing your project. Along the way you will have encountered difficulties which might have modified your original plan so that you can come up with an actual developed project. You can discuss some of these that led to significant changes (if they exist). It is important here is that there is a clear rationale for design and implementation decisions made.

Chapter 5

Testing and Evaluation

The goal of this chapter is an objective evaluation of the final system. The evaluation must be quantitative and not qualitative. Indicate how the system was verified/tested. You may perform qualitative evaluation but this should not form the basis of the main conclusions you derive from the evaluation. This evaluation, where possible, should be comparative, i.e. you should evaluate your system against a commercially available system and/or system detailed in a research publication. If no implementation was done, you could describe the types of experiments/simulations that were carried out. Explain why these and not others were performed and discuss the most important parameters in the tests/simulations and describe their roles.

You should demonstrate operational testing of the project using real or contrived data sets to evaluate aspects of the project not encompassed in the software testing (e.g. quantify how well does your project achieved the overall goal).

- For software based projects this will include, but should not be limited to, evaluation of non-functional requirements.
- For infrastructural projects this testing should include system/network KPI analysis.
- For analysis based projects (ML, malware or other) this may include model evaluation or YARA rule validation, for example.
- For management projects, where software testing or infrastructure testing may not be in scope, the test process for the system is expected to be more rigorous and well described than a project incorporating significant development work.

Summarize the output data, and the statistical or other techniques to deduce your results. Summarize your results, including tables or graphs as appropriate with a brief

description of each. Here, where possible, compare your results with other products/systems. Identify any possible threats to the validity of your results, and discuss each briefly here (you will discuss in more detail in the next chapter).

Chapter 6

Discussion and Conclusions

Write a critical evaluation of your work. You should state your goals and research questions again and outline in short how they were or were not met or answered. What have you achieved? What more could have been accomplished? Mention any problems that affected the progress of your work. Is there any scope for extension/improvement/scalability? What are the new challenges to be faced in the future? The expectation here is that you should discuss the results presented in the previous evaluation section of the project in their totality (i.e. as a whole) from which you will then draw clear conclusions both on the quantitative and qualitative aspects of the overall project. The conclusion section of this chapter should conclude the project.

6.1 Discussion

Discuss how well your solution solves the problem, based on your results from the evaluation chapter. Discuss how well you addressed the project, and what you might do differently if you were to do it again. Make sure to identify how you handled any problems that arose during the project.

6.2 Conclusion

Enumerate the main conclusions you have got in terms of background, problem description and the solution approach you have come up with. Detail your primary and any secondary conclusions from your project.

6.3 Future Work

Discuss any proposals for completion of the project, or for enhancements, or for re-design of your solution or software. Enumerate all the things you would have wanted to do should you have more time to work on this project.

Bibliography

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Appendix A

Code Snippets

Put appendix material in this section e.g. code snippets

If there is too much code to include in the Appendix, submit the rest on a USB disk or CD. Some code should be easily available for examiners to assess.

USE THE APPENDICES

Appendix B

Wireframe Models